



Attacking Autonomic Networks

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#sh run

- Security Analyst @ ERNW GmbH
- Network security and reverse-engineering
- Bachelor and Masters theses are done on Autonomic systems

--More--

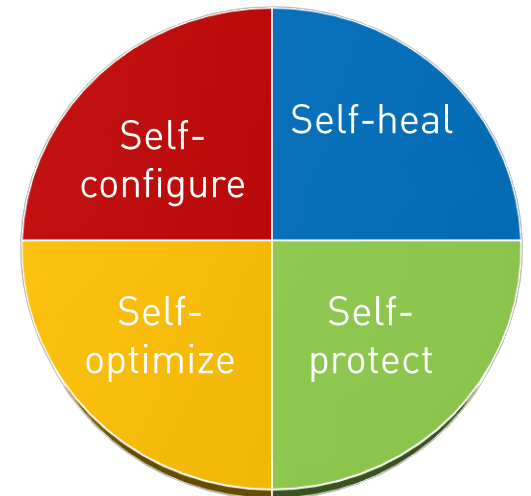
Agenda

- Autonomic Systems
- Cisco deployment of the Autonomic Network
- Reverse-engineer the proprietary protocol
- Discover and exploit multiple vulnerabilities



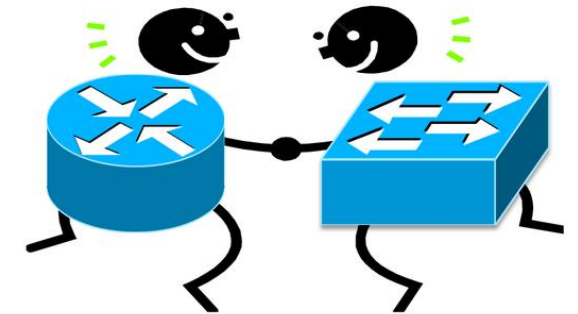
Autonomic Systems

- Smart systems that don't need human intervention to operate
- They have the ability to “self-manage”



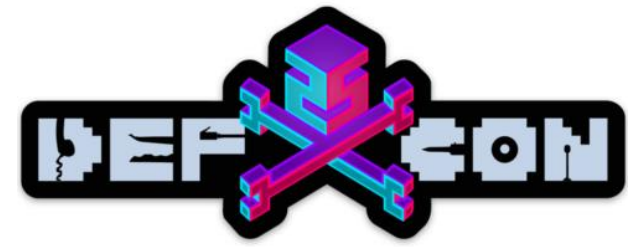
Autonomic Network

- IETF ANIMA working group
- One device that configures everything else
- Only 5 commands are needed
- Nothing has to be configured on the new devices



**Autonomic
Networking**

Autonomic Network logo as shown by Cisco
in their presentations [here](#) and [here](#)



Live Demo

Demo Results

- Plug and Play
- There is no need to configure any command on the greenfield devices
- Only a single command needs to be configured on the brownfield devices

Cisco Deployment

- Communication is divided into 3 phases:
 - Channel Discovery
 - Adjacency Discovery
 - Secure Channel

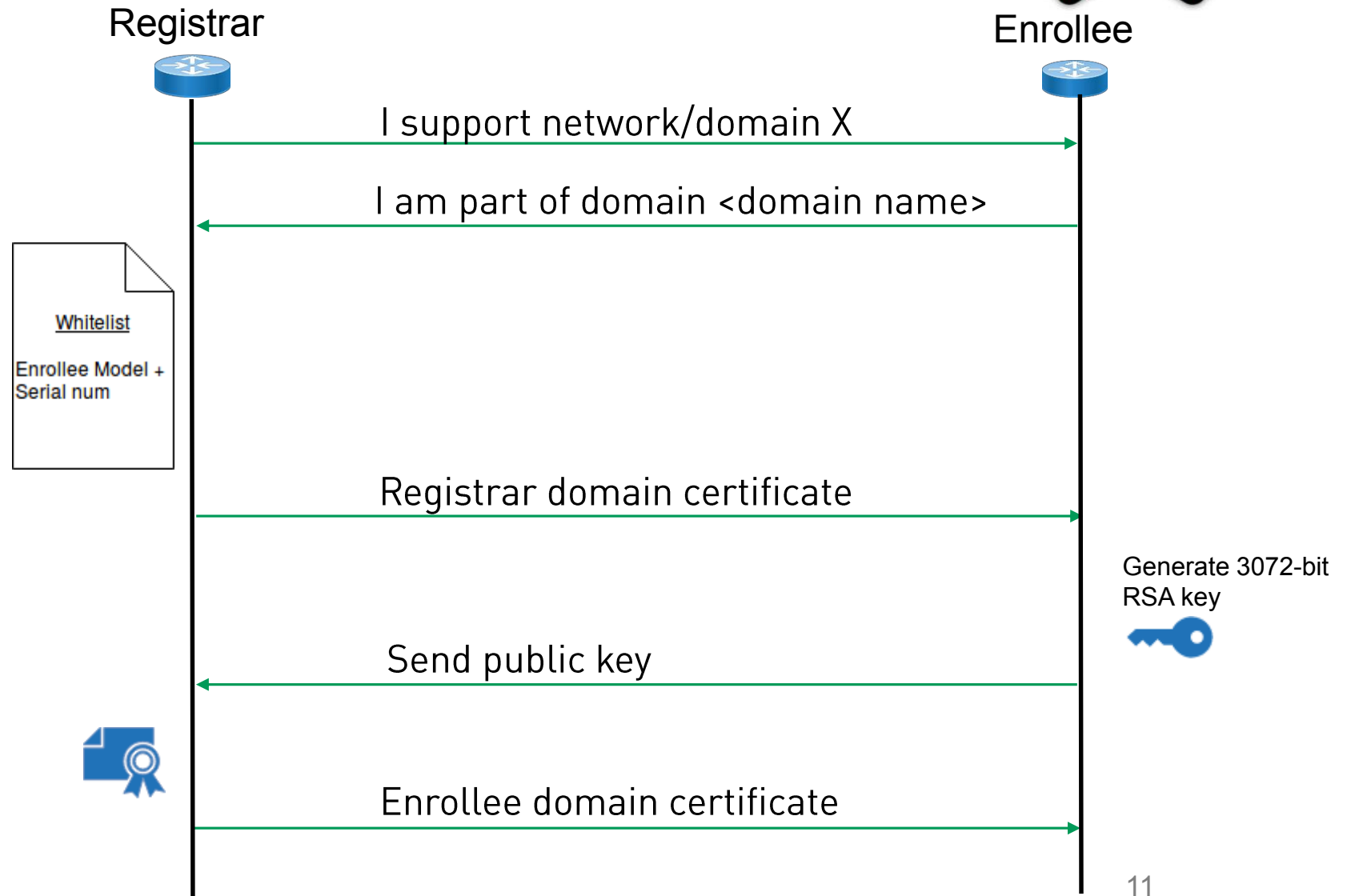
Channel Discovery

- Discover any nearby autonomic devices
- Layer 2 probes sent by registrar

Adjacency Discovery

- Domain name
- Are you allowed to join the domain or not?
 - Rejected: stay at channel discovery phase
 - Allowed: let's issue a certificate then
- UDP port 4936

Adjacency Discovery



Secure Channel

- IPSec
 - Port 500
 - Backwards compatibility
- DIKE
 - Data Internet Key Exchange
 - Port 5000
 - Preferred over IPsec

Registrar Configuration

```
autonomic registrar  
domain-id ERNW.de  
whitelist flash:whitelist.txt  
CA local  
no shut  
autonomic
```

Enrollee Needed Configuration

- Brand new (i.e. no configuration file exists)
 - None!
- Configuration file exists
 - `autonomic`

Autonomic Effect

- IPv6 address based on the domain name and device ID
- Domain Certificate
- VRF cisco_autonomic
- Virtual Interface, ANI1
- Tunnel Interface, Tunnel100000
- AAA (Authentication, Authorization and Accounting) will be enabled
- RADIUS, TFTP, Syslog (if available)

Are you in Control?

Autonomic Network: Under The Hood

Apply a display filter ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	00:62:ec:9d:80:60	ISL-Frame_cd:cd:dc	LLC	118	U, func=UI; SNAP, OUI 0x00000C (Cisco), PID 0x88EF
2	0.014104	00:00:00_00:00:01	ISL-Frame_cd:cd:dc	LLC	148	U, func=UI; SNAP, OUI 0x00000C (Cisco), PID 0x88EF
3	11.680271	00:00:00_00:00:01	ISL-Frame_cd:cd:dc	LLC	159	U, func=UI; SNAP, OUI 0x00000C (Cisco), PID 0x88EF
4	21.678386	00:62:ec:9d:80:60	ISL-Frame_cd:cd:dc	LLC	212	U, func=UI; SNAP, OUI 0x00000C (Cisco), PID 0x88EF
5	21.678411	00:62:ec:9d:80:60	ISL-Frame_cd:cd:dc	LLC	148	U, func=UI; SNAP, OUI 0x00000C (Cisco), PID 0x88EF
6	24.384456	00:62:ec:9d:80:60	ISL-Frame_cd:cd:dc	LLC	1436	U, func=UI; SNAP, OUI 0x00000C (Cisco), PID 0x88EF
7	24.384480	00:62:ec:9d:80:60	ISL-Frame_cd:cd:dc	LLC	1365	U, func=UI; SNAP, OUI 0x00000C (Cisco), PID 0x88EF
8	24.506508	00:00:00_00:00:01	ISL-Frame_cd:cd:dc	LLC	1436	U, func=UI; SNAP, OUI 0x00000C (Cisco), PID 0x88EF
9	24.506526	00:00:00_00:00:01	ISL-Frame_cd:cd:dc	LLC	153	U, func=UI; SNAP, OUI 0x00000C (Cisco), PID 0x88EF
10	26.502154	00:62:ec:9d:80:60	ISL-Frame_cd:cd:dc	LLC	1213	U, func=UI; SNAP, OUI 0x00000C (Cisco), PID 0x88EF
11	28.727965	00:62:ec:9d:80:60	ISL-Frame_cd:cd:dc	LLC	596	U, func=UI; SNAP, OUI 0x00000C (Cisco), PID 0x88EF
12	30.621816	00:62:ec:9d:80:60	ISL-Frame_cd:cd:dc	LLC	596	U, func=UI; SNAP, OUI 0x00000C (Cisco), PID 0x88EF

Channel Discovery

6 bytes	6 bytes	2 bytes	Till 1500 bytes	
Destination MAC Address	Source MAC Address	EtherType	Payload	FCS

Ethernet II

6 bytes	6 bytes	2 bytes	1 byte	1 byte	1 byte	Till 1500 bytes	
Destination MAC Address	Source MAC Address	Frame Length	DSAP	SSAP	Control	Payload	FCS

802.3
(802.3, 802.2 LLC)

6 bytes	6 bytes	2 bytes	1 byte	1 byte	1 byte	3 bytes	2 bytes	Till 1500 bytes	
Destination MAC Address	Source MAC Address	Frame Length	DSAP	SSAP	Control	OUI	Protocol ID	Payload	FCS

802.3
(802.3, 802.2 SNAP)

Not Ethernet II

SNAP Frame

	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	
0000	01	00	0c	cd	cd	dc	00	62	ec	9d	80	60	00	68	aa	aa	b...`h..
0010	03	00	00	0c	88	ef	10	01	00	ff	00	01	00	60	00	00	`..
0020	00	00	01	00	00	1e	50	49	44	3a	49	53	52	34	33	32	PID:ISR432
0030	31	2f	4b	39	20	53	4e	3a	46	44	4f	32	30	31	38	41	1/K9 SN:FD02018A
0040	30	4d	38	00	02	00	00	14	47	69	67	61	62	69	74	45	0M8.....GigabitE
0050	74	68	65	72	6e	65	74	30	2f	30	2f	30	03	00	00	00	thernet0/0/0....
0060	04	00	00	02	00	00	05	00	00	04	00	00	00	00	06	00	
0070	00	04	00	00	00	08											

Channel Discovery

Ethernet

	Destination MAC Address						Source MAC Address						Frame Length		SNAP Frame	
	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
0000	01	00	0c	cd	cd	dc	00	62	ec	9d	80	60	00	68	aa	aa
0010	03	00	00	0c	88	ef	10	01	00	ff	00	01	00	60	00	00
0020	00	00	01	00	00	1e	50	49	44	3a	49	53	52	34	33	32
0030	31	2f	4b	39	20	53	4e	3a	46	44	4f	32	30	31	38	41
0040	30	4d	38	00	02	00	00	14	47	69	67	61	62	69	74	45
0050	74	68	65	72	6e	65	74	30	2f	30	2f	30	03	00	00	00
0060	04	00	00	02	00	00	05	00	00	04	00	00	00	00	06	00
0070	00	04	00	00	00	08										

Channel Discovery

Ethernet

Organization Unique Identifier

AN Protocol ID

	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	
0000	01	00	0c	cd	cd	dc	00	62	ec	9d	80	60	00	68	aa	aa	b...`h..
0010	03	00	00	0c	88	ef	10	01	00	ff	00	01	00	60	00	00	`..
0020	00	00	01	00	00	1e	50	49	44	3a	49	53	52	34	33	32	PID:ISR432
0030	31	2f	4b	39	20	53	4e	3a	46	44	4f	32	30	31	38	41	1/K9 SN:FD02018A
0040	30	4d	38	00	02	00	00	14	47	69	67	61	62	69	74	45	0M8.....GigabitE
0050	74	68	65	72	6e	65	74	30	2f	30	2f	30	03	00	00	00	thernet0/0/0....
0060	04	00	00	02	00	00	05	00	00	04	00	00	00	00	06	00	
0070	00	04	00	00	00	08											

Channel Discovery

Channel Discovery

Ethernet

Octet	0								1								2								3							
Bits	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
32	Version				Reserved				State								Factory Default															
64	Operation Code																Length															
96	Reserved																															
128	TLV (Options)																															

AN Channel Discovery Header

Channel Discovery

Channel Discovery

Ethernet

Version = 1, reserved = 0						State				Factory Default						Operation Code	
	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	
0000	01	00	0c	cd	cd	dc	00	62	ec	9d	80	60	00	68	aa	aa	b...`.h..
0010	03	00	00	0c	88	ef	10	01	00	ff	00	01	00	60	00	00	`. ..
0020	00	00	01	00	00	1e	50	49	44	3a	49	53	52	34	33	32	PID:ISR432
0030	31	2f	4b	39	20	53	4e	3a	46	44	4f	32	30	31	38	41	1/K9 SN:FD02018A
0040	30	4d	38	00	02	00	00	14	47	69	67	61	62	69	74	45	0M8.....GigabitE
0050	74	68	65	72	6e	65	74	30	2f	30	2f	30	03	00	00	00	thernet0/0/0....
0060	04	00	00	02	00	00	05	00	00	04	00	00	00	00	06	00	
0070	00	04	00	00	00	08											

Channel Discovery

Channel Discovery

Ethernet

Opcode Value	Significance
0x0001	Registrar/Enrollee announcement
0x0002	Receiver reply for the announcement
0x0003	Sender acknowledgment for the reply
0x0004	Keepalive probes

Channel Discovery

Channel Discovery

Ethernet

	Header Length				Reserved				Type				Length				
	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	
0000	01	00	0c	cd	cd	dc	00	62	ec	9d	80	60	00	68	aa	aa	b...`h..
0010	03	00	00	0c	88	ef	10	01	00	ff	00	01	00	60	00	00	`..
0020	00	00	01	00	00	1e	50	49	44	3a	49	53	52	34	33	32	PID:ISR432
0030	31	2f	4b	39	20	53	4e	3a	46	44	4f	32	30	31	38	41	1/K9 SN:FD02018A
0040	30	4d	38	00	02	00	00	14	47	69	67	61	62	69	74	45	0M8.....GigabitE
0050	74	68	65	72	6e	65	74	30	2f	30	2f	30	03	00	00	00	thernet0/0/0....
0060	04	00	00	02	00	00	05	00	00	04	00	00	00	00	06	00	
0070	00	04	00	00	00	08											

Channel Discovery

Channel Discovery

Ethernet

Option Type	Significance
0x0100	Source UDI
0x0200	Source Interface
0x0300	Receiver UDI
0x0400	2 bytes of zeros
0x0500	4 bytes of zeros
0x0600	4 bytes of value 0x00000008

Adjacency Discovery

	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	
0000	01	00	0c	cd	cd	dc	00	62	ec	9d	80	60	00	c6	aa	aa	b...`....
0001	03	00	00	0c	88	ef	10	05	00	ff	00	00	00	0e	00	00	
0002	00	00	86	dd	60	00	00	00	00	88	11	ff	fe	80	00	00	`.....
0003	00	00	00	00	02	62	ec	ff	fe	9d	80	60	ff	02	00	00	b.....`....
0004	00	00	00	00	00	00	00	00	00	00	01	50	13	48	13	48	P.H.H
0005	00	88	86	00	20	02	00	ff	00	01	00	80	00	00	00	00	
0006	00	01	00	22	50	49	44	3a	49	53	52	34	33	32	31	2f	... "PID:ISR4321/
0007	4b	39	20	53	4e	3a	46	44	4f	32	30	31	38	41	30	4d	K9 SN:FD02018A0M
0008	38	00	00	02	00	15	30	30	36	32	2e	65	63	39	64	2e	8.....0062.ec9d.
0009	38	30	36	30	2d	31	00	00	03	00	0c	45	52	4e	57	2e	8060-1.....ERNW.
0010	64	65	00	00	07	00	14	fe	80	00	00	00	00	00	00	02	de.....
0011	62	ec	ff	fe	9d	80	60	00	08	00	09	41	4e	49	31	00	b.....`.....ANI1.
0012	00	05	00	14	fd	b6	67	6a	9a	78	00	00	00	62	ec	9d	gj.x...b..
0013	80	60	00	01													

Adjacency Discovery

Ethernet

	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	
0000	01	00	0c	cd	cd	dc	00	62	ec	9d	80	60	00	c6	aa	aa	b....`....
0001	03	00	00	0c	88	ef	10	05	00	ff	00	00	00	0e	00	00	
0002	00	00	86	dd	60	00	00	00	00	88	11	ff	fe	80	00	00	`.....
0003	00	00	00	00	02	62	ec	ff	fe	9d	80	60	ff	02	00	00	b.....`....
0004	00	00	00	00	00	00	00	00	00	00	01	50	13	48	13	48	P.H.H
0005	00	88	86	00	20	02	00	ff	00	01	00	80	00	00	00	00	
0006	00	01	00	22	50	49	44	3a	49	53	52	34	33	32	31	2f	... "PID:ISR4321/
0007	4b	39	20	53	4e	3a	46	44	4f	32	30	31	38	41	30	4d	K9 SN:FD02018A0M
0008	38	00	00	02	00	15	30	30	36	32	2e	65	63	39	64	2e	8.....0062.ec9d.
0009	38	30	36	30	2d	31	00	00	03	00	0c	45	52	4e	57	2e	8060-1.....ERNW.
0010	64	65	00	00	07	00	14	fe	80	00	00	00	00	00	00	02	de.....
0011	62	ec	ff	fe	9d	80	60	00	08	00	09	41	4e	49	31	00	b.....`.....ANI1.
0012	00	05	00	14	fd	b6	67	6a	9a	78	00	00	00	62	ec	9d	gj.x...b..
0013	80	60	00	01													

Ethernet 802.3 /802.2 SNAP

Adjacency Discovery

Customized CD Header

Ethernet

	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	
0000	01	00	0c	cd	cd	dc	00	62	ec	9d	80	60	00	c6	aa	aa	b....`....
0001	03	00	00	0c	88	ef	10	05	00	ff	00	00	00	0e	00	00	
0002	00	00	86	dd	60	00	00	00	00	88	11	ff	fe	80	00	00	`.....
0003	00	00	00	00	02	62	ec	ff	fe	9d	80	60	ff	02	00	00	b.....`....
0004	00	00	00	00	00	00	00	00	00	00	01	50	13	48	13	48	P.H.H
0005	00	88	86	00	20	02	00	ff	00	01	00	80	00	00	00	00	
0006	00	01	00	22	50	49	44	3a	49	53	52	34	33	32	31	2f	... "PID:ISR4321/
0007	4b	39	20	53	4e	3a	46	44	4f	32	30	31	38	41	30	4d	K9 SN:FD02018A0M
0008	38	00	00	02	00	15	30	30	36	32	2e	65	63	39	64	2e	8.....0062.ec9d.
0009	38	30	36	30	2d	31	00	00	03	00	0c	45	52	4e	57	2e	8060-1.....ERNW.
0010	64	65	00	00	07	00	14	fe	80	00	00	00	00	00	00	02	de.....
0011	62	ec	ff	fe	9d	80	60	00	08	00	09	41	4e	49	31	00	b.....`.....ANI1.
0012	00	05	00	14	fd	b6	67	6a	9a	78	00	00	00	62	ec	9d	gj.x...b..
0013	80	60	00	01													

Customized CD Header

Adjacency Discovery

CD Header Field	Value (hex)
Version	1
Reserved	0
State	05
Factory Default	00 ff
Operation Code	00
Length	0e
Reserved	00 00 00 00
Ethertype	86 dd

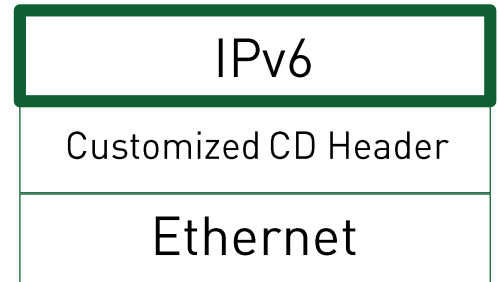
```

00 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15
0000  01 00 0c cd cd dc 00 62 ec 9d 80 60 00 c6 aa aa .....b...`....
0001  03 00 00 0c 88 ef 10 05 00 ff 00 00 00 0e 00 00 .....
0002  00 00 86 dd 60 00 00 00 00 88 11 ff fe 80 00 00 .....
0003  00 00 00 00 02 62 ec ff fe 9d 80 60 ff 02 00 00 .....b.....`....
0004  00 00 00 00 00 00 00 00 00 00 01 50 13 48 13 48 .....P.H.H
0005  00 88 86 00 20 02 00 ff 00 01 00 80 00 00 00 00 .....
0006  00 01 00 22 50 49 44 3a 49 53 52 34 33 32 31 2f ... "PID:ISR4321/
0007  4b 39 20 53 4e 3a 46 44 4f 32 30 31 38 41 30 4d K9 SN:FD02018A0M
0008  38 00 00 02 00 15 30 30 36 32 2e 65 63 39 64 2e 8.....0062.ec9d.
0009  38 30 36 30 2d 31 00 00 03 00 0c 45 52 4e 57 2e 8060-1.....ERNW.
0010  64 65 00 00 07 00 14 fe 80 00 00 00 00 00 00 02 de.....
0011  62 ec ff fe 9d 80 60 00 08 00 09 41 4e 49 31 00 b.....`....ANI1.
0012  00 05 00 14 fd b6 67 6a 9a 78 00 00 00 62 ec 9d .....gj.x...b..
0013  80 60 00 01

```

Customized CD Header

Adjacency Discovery



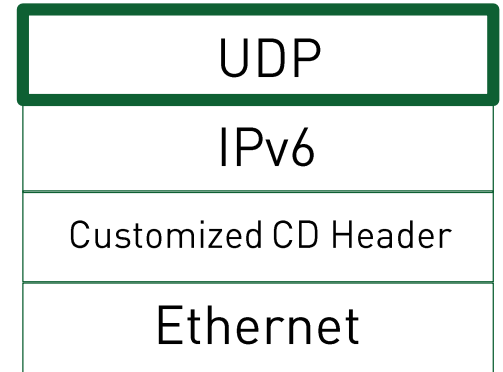
```

00 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15
0000  01 00 0c cd cd dc 00 62 ec 9d 80 60 00 c6 aa aa .....b...`....
0001  03 00 00 0c 88 ef 10 05 00 ff 00 00 00 0e 00 00 .....
0002  00 00 86 dd 60 00 00 00 00 88 11 ff fe 80 00 00 .....
0003  00 00 00 00 02 62 ec ff fe 9d 80 60 ff 02 00 00 .....b.....`....
0004  00 00 00 00 00 00 00 00 00 00 01 50 13 48 13 48 .....P.H.H
0005  00 88 86 00 20 02 00 ff 00 01 00 80 00 00 00 00 .....
0006  00 01 00 22 50 49 44 3a 49 53 52 34 33 32 31 2f ..."PID:ISR4321/
0007  4b 39 20 53 4e 3a 46 44 4f 32 30 31 38 41 30 4d K9 SN:FD02018A0M
0008  38 00 00 02 00 15 30 30 36 32 2e 65 63 39 64 2e 8.....0062.ec9d.
0009  38 30 36 30 2d 31 00 00 03 00 0c 45 52 4e 57 2e 8060-1.....ERNW.
0010  64 65 00 00 07 00 14 fe 80 00 00 00 00 00 00 02 de.....
0011  62 ec ff fe 9d 80 60 00 08 00 09 41 4e 49 31 00 b.....`....ANI1.
0012  00 05 00 14 fd b6 67 6a 9a 78 00 00 00 62 ec 9d .....gj.x...b..
0013  80 60 00 01

```

IPv6 Header

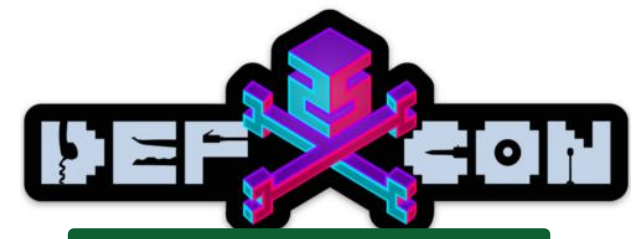
Adjacency Discovery



```

00 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15
0000  01 00 0c cd cd dc 00 62 ec 9d 80 60 00 c6 aa aa .....b...`....
0001  03 00 00 0c 88 ef 10 05 00 ff 00 00 00 0e 00 00 .....
0002  00 00 86 dd 60 00 00 00 00 88 11 ff fe 80 00 00 .....
0003  00 00 00 00 02 62 ec ff fe 9d 80 60 ff 02 00 00 .....b.....`....
0004  00 00 00 00 00 00 00 00 00 00 01 50 13 48 13 48 .....P.H.H
0005  00 88 86 00 20 02 00 ff 00 01 00 80 00 00 00 00 .....
0006  00 01 00 22 50 49 44 3a 49 53 52 34 33 32 31 2f ..."PID:ISR4321/
0007  4b 39 20 53 4e 3a 46 44 4f 32 30 31 38 41 30 4d K9 SN:FD02018A0M
0008  38 00 00 02 00 15 30 30 36 32 2e 65 63 39 64 2e 8.....0062.ec9d.
0009  38 30 36 30 2d 31 00 00 03 00 0c 45 52 4e 57 2e 8060-1.....ERNW.
0010  64 65 00 00 07 00 14 fe 80 00 00 00 00 00 00 02 de.....
0011  62 ec ff fe 9d 80 60 00 08 00 09 41 4e 49 31 00 b.....`....ANI1.
0012  00 05 00 14 fd b6 67 6a 9a 78 00 00 00 62 ec 9d .....gj.x...b..
0013  80 60 00 01
  
```

UDP Header



Adjacency Discovery

AD Header

UDP

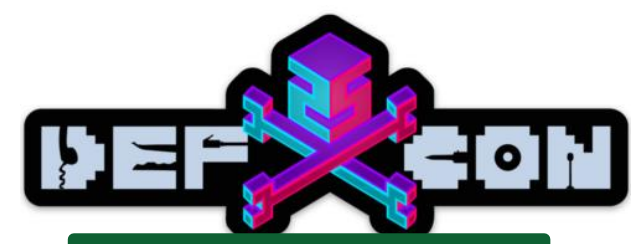
IPv6

Customized CD Header

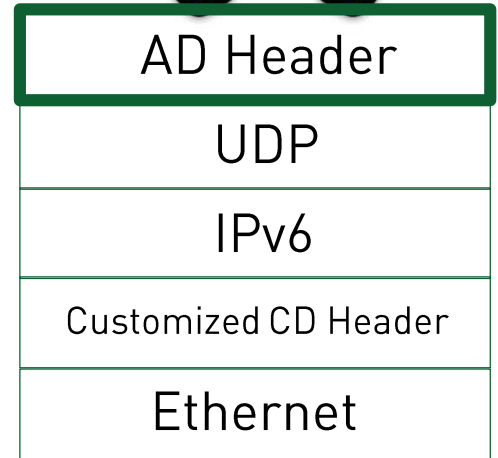
Ethernet

Octet	0								1								2								3							
Bits	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
32	Version				Reserved				State								Factory Default															
64	Operation Code																Length															
96	Reserved																															
128	TLV (Options)																															

AN Adjacency Discovery Header



Adjacency Discovery



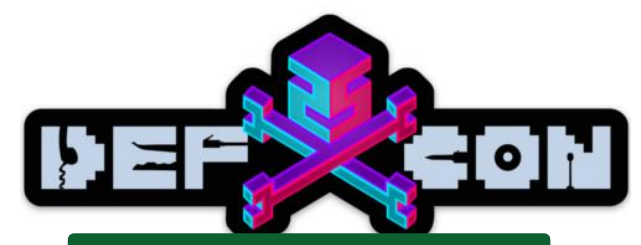
```

00 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15
0000 01 00 0c cd cd dc 00 62 ec 9d 80 60 00 c6 aa aa .....b...`....
0001 03 00 00 0c 88 ef 10 05 00 ff 00 00 00 0e 00 00 .....
0002 00 00 86 dd 60 00 00 00 00 88 11 ff fe 80 00 00 ....`.....
0003 00 00 00 00 02 62 ec ff fe 9d 80 60 ff 02 00 00 .....b.....`....
0004 00 00 00 00 00 00 00 00 00 00 01 50 13 48 13 48 .....P.H.H
0005 00 88 86 00 20 02 00 ff 00 01 00 80 00 00 00 00 .....
0006 00 01 00 22 50 49 44 3a 49 53 52 34 33 32 31 2f ... "PID:ISR4321/
0007 4b 39 20 53 4e 3a 46 44 4f 32 30 31 38 41 30 4d K9 SN:FD02018A0M
0008 38 00 00 02 00 15 30 30 36 32 2e 65 63 39 64 2e 8.....0062.ec9d.
0009 38 30 36 30 2d 31 00 00 03 00 0c 45 52 4e 57 2e 8060-1.....ERNW.
0010 64 65 00 00 07 00 14 fe 80 00 00 00 00 00 00 02 de.....
0011 62 ec ff fe 9d 80 60 00 08 00 09 41 4e 49 31 00 b.....`....ANI1.
0012 00 05 00 14 fd b6 67 6a 9a 78 00 00 00 62 ec 9d .....gj.x...b..
0013 80 60 00 01

```

Version = 2, reserved = 0

State



Adjacency Discovery

AD Header

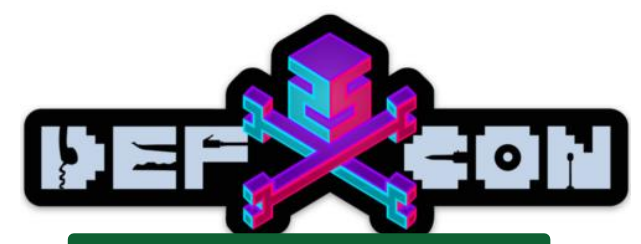
UDP

IPv6

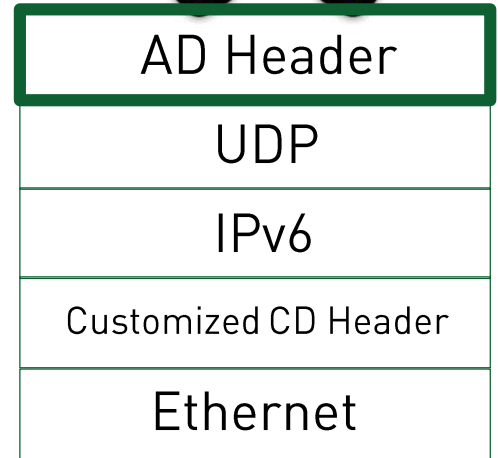
Customized CD Header

Ethernet

State Value	Significance
0x02	Multicast, Neighbor Discovery hello packets
0x03	Unicast, Bootstrap phase
0x04	Unicast, negotiating secure channel parameters



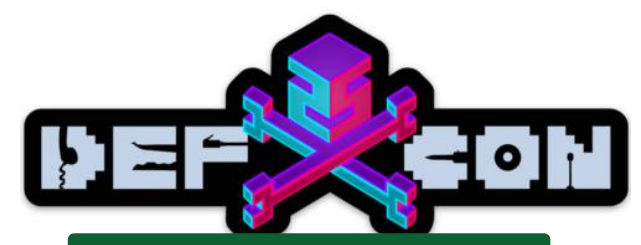
Adjacency Discovery



	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	
0000	01	00	0c	cd	cd	dc	00	62	ec	9d	80	60	00	c6	aa	aa	b...`....
0001	03	00	00	0c	88	ef	10	05	00	ff	00	00	00	0e	00	00	
0002	00	00	86	dd	60	00	00	00	88	11	ff	fe	80	00	00	00	`.....
0003	00	00	00	00	02	62	ec	ff	fe	9d	80	60	ff	02	00	00	b.....`....
0004	00	00	00	00	00	00	00	00	00	00	01	50	13	48	13	48	P.H.H
0005	00	88	86	00	20	02	00	ff	00	01	00	80	00	00	00	00	
0006	00	01	00	22	50	49	44	3a	49	53	52	34	33	32	31	2f	... "PID:ISR4321/
0007	4b	39	20	53	4e	3a	46	44	4f	32	30	31	38	41	30	4d	K9 SN:FD02018A0M
0008	38	00	00	02	00	15	30	30	36	32	2e	65	63	39	64	2e	8.....0062.ec9d.
0009	38	30	36	30	2d	31	00	00	03	00	0c	45	52	4e	57	2e	8060-1.....ERNW.
0010	64	65	00	00	07	00	14	fe	80	00	00	00	00	00	00	02	de.....
0011	62	ec	ff	fe	9d	80	60	00	08	00	09	41	4e	49	31	00	b.....`.....ANI1.
0012	00	05	00	14	fd	b6	67	6a	9a	78	00	00	00	62	ec	9d	gj.x...b..
0013	80	60	00	01													

Reserved

Operation Code



Adjacency Discovery

AD Header

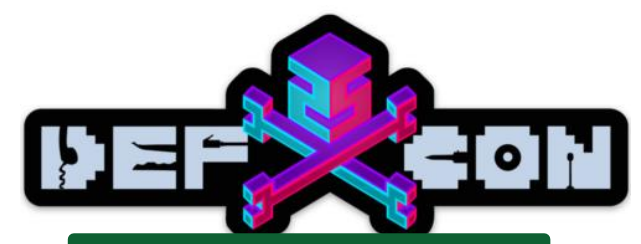
UDP

IPv6

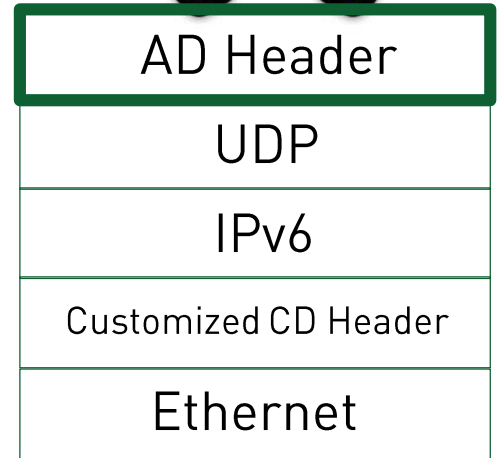
Customized CD Header

Ethernet

Opcode Value	Significance
0x0001	Neighbor Discovery Domain packets
0x0003	Whitelist acceptance/rejection for the requesting nodes
0x0004	Device Domain Certificate
0x0005	Bootstrap invite by the registrar
0x0007	Bootstrap reply by the enrollee
0x0008	Device Domain Certificate (rarely used)
0x0019	Negotiating available security parameters for the secure channel
0x001a	Acknowledgment on the agreed security parameters
0x001c	Failed to build the secure channel



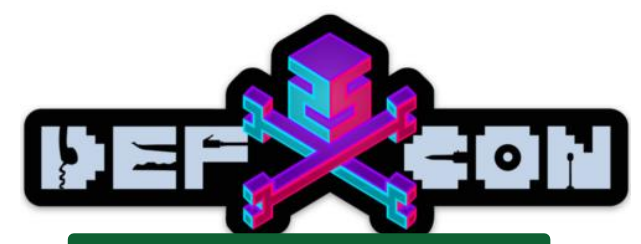
Adjacency Discovery



	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	
0000	01	00	0c	cd	cd	dc	00	62	ec	9d	80	60	00	c6	aa	aa	b....`.....
0001	03	00	00	0c	88	ef	10	05	00	ff	00	00	00	0e	00	00	
0002	00	00	86	dd	60	00	00	00	00	88	11	ff	fe	80	00	00	`.....
0003	00	00	00	00	02	62	ec	ff	fe	9d	80	60	ff	02	00	00	b.....`.....
0004	00	00	00	00	00	00	00	00	00	00	01	50	13	48	13	48	P.H.H
0005	00	88	86	00	20	02	00	ff	00	01	00	80	00	00	00	00	
0006	00	01	00	22	50	49	44	3a	49	53	52	34	33	32	31	2f	... "PID:ISR4321/
0007	4b	39	20	53	4e	3a	46	44	4f	32	30	31	38	41	30	4d	K9 SN:FD02018A0M
0008	38	00	00	02	00	15	30	30	36	32	2e	65	63	39	64	2e	8.....0062.ec9d.
0009	38	30	36	30	2d	31	00	00	03	00	0c	45	52	4e	57	2e	8060-1.....ERNW.
0010	64	65	00	00	07	00	14	fe	80	00	00	00	00	00	00	02	de.....
0011	62	ec	ff	fe	9d	80	60	00	08	00	09	41	4e	49	31	00	b.....`.....ANI1.
0012	00	05	00	14	fd	b6	67	6a	9a	78	00	00	00	62	ec	9d	gj.x...b..
0013	80	60	00	01													

Header Length

Factory Default



Adjacency Discovery

AD Header

UDP

IPv6

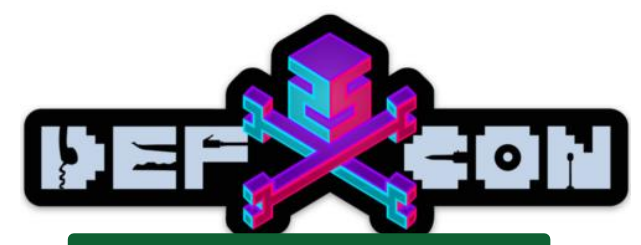
Customized CD Header

Ethernet

	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	
0000	01	00	0c	cd	cd	dc	00	62	ec	9d	80	60	00	c6	aa	aa	b....`....
0001	03	00	00	0c	88	ef	10	05	00	ff	00	00	00	0e	00	00	
0002	00	00	86	dd	60	00	00	00	88	11	ff	fe	80	00	00	00	`.....
0003	00	00	00	00	02	62	ec	ff	fe	9d	80	60	ff	02	00	00	b.....`....
0004	00	00	00	00	00	00	00	00	00	00	01	50	13	48	13	48	P.H.H
0005	00	88	86	00	20	02	00	ff	00	01	00	80	00	00	00	00	
0006	00	01	00	22	50	49	44	3a	49	53	52	34	33	32	31	2f	... "PID:ISR4321/
0007	4b	39	20	53	4e	3a	46	44	4f	32	30	31	38	41	30	4d	K9 SN:FD02018A0M
0008	38	00	00	02	00	15	30	30	36	32	2e	65	63	39	64	2e	8.....0062.ec9d.
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0010	64	65	00	00	07	00	14	fe	80	00	00	00	00	00	00	02	de.....
0011	62	ec	ff	fe	9d	80	60	00	08	00	09	41	4e	49	31	00	b.....`.....ANI1.
0012	00	05	00	14	fd	b6	67	6a	9a	78	00	00	00	62	ec	9d	gj.x...b..
0013	80	60	00	01													

Type

Length



Adjacency Discovery

AD Header

UDP

IPv6

Customized CD Header

Ethernet

Operation Codes	Available field types	Fields Significance
0x0001	0x0001	Source UDI
	0x0002	Source Device Domain ID
	0x0003	Domain Name
⋮	⋮	⋮
0x0019	0x0001	Security Channel Protection Mode, either DIKE or IPSEC
0x001a	0x0001	Acknowledgment on the agreed Security Mode
0x001c	0x0001	Failed to build the Secure Channel

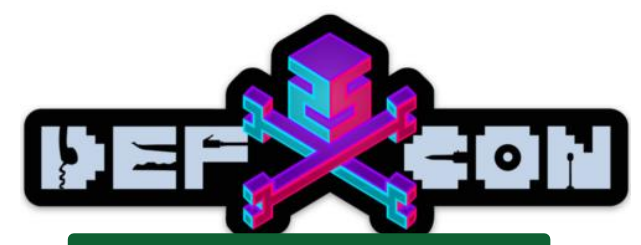
Secure Channel

734 2017-02-01 10:44:01.135104828 fe80::f27f:6ff:fee2:6450 fe80::262:ecff:fe9d:8060 UDP 184 5000→5000 Len...

Wireshark · Packet 734 · full flow

- ▶ Frame 734: 184 bytes on wire (1472 bits), 184 bytes captured (1472 bits) on interface 0
- ▶ IEEE 802.3 Ethernet
- ▶ Logical-Link Control
- ▶ Ethernet II, Src: Atrie_00:00:00 (00:0e:00:00:00:00), Dst: 10:05:00:ff:00:00 (10:05:00:ff:00:00)
- ▶ Internet Protocol Version 6, Src: fe80::f27f:6ff:fee2:6450, Dst: fe80::262:ecff:fe9d:8060
- ▶ User Datagram Protocol, Src Port: 5000, Dst Port: 5000
- ▶ Data (100 bytes)

0000	01 00 0c cd cd dc f0 7f	06 e2 64 50 00 aa aa aa	dP....
0010	03 00 00 0c 88 ef 10 05	00 ff 00 00 00 0e 00 00	
0020	00 00 86 dd 6c 00 00 00	00 6c 11 40 fe 80 00 00	l... .l.@....
0030	00 00 00 00 f2 7f 06 ff	fe e2 64 50 fe 80 00 00	dP....
0040	00 00 00 00 02 62 ec ff	fe 9d 80 60 13 88 13 88	b.. ...`....
0050	00 6c 5a 10 00 00 00 00	40 ca 48 c4 63 2c f1 fe	.lZ..... @.H.c,..
0060	36 49 57 d7 ec f9 49 97	2e 20 25 20 00 00 00 02	6IW...I. . %
0070	00 00 00 60 00 00 00 44	41 2d f5 5a 29 57 50 80	...`...D A-.Z)WP.
0080	a9 41 39 75 69 ae 91 98	8e d6 82 f4 35 76 87 da	.A9ui...5v..
0090	e4 81 c8 19 f2 4c 09 56	e8 3a f4 3a 6f 75 0f 9f	L.V ...:ou..
00a0	fa a5 4c 83 11 fa 2c e9	7b c4 d7 7a fe f2 ba ef	..L..., {..Z....
00b0	21 ce c9 cb d7 e3 06 a1		!.....



Secure Channel

- Supports AN since 2014
- DIKE only supported on newer operating Systems
- IPSec NULL 😊

Secure Channel

UDP

IPv6

Customized CD Header

Ethernet

ME 3600X-24CX-M



1567 737.585685497 fe80::46e4:d9ff:fe9b:979c ff02::2 ICMPv6 150 RPL Control (DODAG Information Obj...

Wireshark · Packet 1567 · wireshark_eth0_20161104102507_F7K11U

- Frame 1567: 150 bytes on wire (1200 bits), 150 bytes captured (1200 bits) on interface 0
- Ethernet II, Src: CiscoInc_9b:97:c4 (44:e4:d9:9b:97:c4), Dst: CadmusCo_b1:bd:bc (08:00:27:b1:bd:bc)
- Internet Protocol Version 6, Src: fe80::46e4:d9ff:fe9b:97c4, Dst: fe80::1
- Encapsulating Security Payload
- Generic Routing Encapsulation (IPv6)
- Internet Protocol Version 6, Src: fe80::46e4:d9ff:fe9b:979c, Dst: ff02::2
- Internet Control Message Protocol v6

0000	08 00 27 b1 bd bc 44 e4 d9 9b 97 c4 86 dd 60 00	.. .D.
0010	00 00 00 60 32 ff fe 80 00 00 00 00 00 46 e4	...`2... ..F.
0020	d9 ff fe 9b 97 c4 fe 80 00 00 00 00 00 00 00	
0030	00 00 00 00 00 01 19 c1 5b f5 00 00 01 38 00 00	 [....8..
0040	86 dd 60 00 00 00 00 1c 3a ff fe 80 00 00 00 00	..`..... :.....
0050	00 00 46 e4 d9 ff fe 9b 97 9c ff 02 00 00 00 00	..F.....
0060	00 00 00 00 00 00 00 00 00 02 9b 01 45 83 00 15	E...
0070	01 00 18 55 00 00 fd 0a 7c 9c df 87 00 00 00 1e	...U....
0080	bd c8 3a 00 00 02 01 02 02 2f 21 99 8d d2 d2 03	 /!.....
0090	60 3f 6e b1 2d a3	`?n.-.

No.: 1567 · Time: 737.585685497 · Source: fe80::46e4:d9ff:fe9b:979c · Destination: ff02::2 · Protocol: ICMPv6 · Length: 150 · Info: RPL Control (DODAG Information Object)

Is it Secure?

Live Chat



Support

Live Chat



Support

Me:

Hi, I connected 2 nodes from 2 different domains and they built the secure channel!

Live Chat



Support

Me:

Hi, I connected 2 nodes from 2 different domains and they built the secure channel!

Support:

Thanks for reporting, we created BugID CSCvd15717. We will check with the BU for that

Live Chat



Support

Me:

Hi, I connected 2 nodes from 2 different domains and they built the secure channel!

Support:

Hi, the BU responded that as both have a certificate signed by same CA, then they can connect.

Live Chat



Support

Me:

Hi, I connected 2 nodes from 2 different domains and they built the secure channel!

Support:

Hi, the BU responded that as both have a certificate signed by same CA, then they can connect.

Me:

Wait, what about different domains? Well, this shouldn't be

Live Chat



Support

Me:

Hi, I connected 2 nodes from 2 different domains and they built the secure channel!

Support:

Hi, the BU responded that as both have a certificate signed by same CA, then they can connect.

Me:

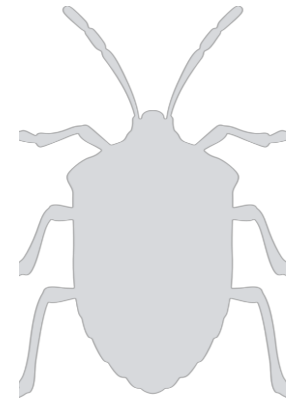
Wait, what about different domains? Well, this shouldn't be

Support:

We will add a feature to check domains in the future!

Bug: CSCvd15717

- Different domains can connect as long as they have certificates from the same CA
- A feature of checking domains will be added in the future
- Whitelist is not checked when the enrollee has a certificate
- No mechanism to stop enrollee with a certificate from joining your domain



Live Chat



Support

Me:

Hi, I can't revoke the certificate of one of the accepted nodes.

Live Chat



Support

Me:

Hi, I can't revoke the certificate of one of the accepted nodes.

Support:

We will check that. Please note that the revoking of certificates is not supported on local CA.

Live Chat



Support

Me:

Hi, I can't revoke the certificate of one of the accepted nodes.

Support:

We created CVE-2017-6664 for that.

CVE-2017-6664

- Certification Revocation List is not correctly implemented on IOS XE
- No way to protect against malicious nodes within the network

Live Chat



Support

Me:

Hi, the attacker can reset remotely the secure channel every time they are created, not only this the information is also in plain text!

Live Chat



Support

Me:

Hi, the attacker can reset remotely the secure channel every time they are created, not only this the information is also in plain text!

Support:

We created CVE-2017-6665 for that.

CVE-2017-6665

- Replaying the Channel Discovery and Adjacency Discovery packets of any of the accepted nodes reset the Secure channel
- Secure channel is vulnerable to denial-of-service attacks
- Once the secure channel resets, the encrypted information is sent in plain text

Live Chat



Support

Me:

Hi, if the attacker reset the channel multiple times, eventually the node crashes!

Live Chat



Support

Me:

Hi, if the attacker reset the channel multiple times, eventually the node crashes down!

Support:

We created CVE-2017-6663 for that.

CVE-2017-6663

- Resetting the secure channel multiple times will crash the nodes due to a problem in how mDNS packets are handled
- It usually takes about 15 minutes to crash the device

Live Chat



Support

Me:

Hi, the attacker can crash the registrar by sending invalid enrollee IDs

Support:

We created CVE-2017-3849 for that.

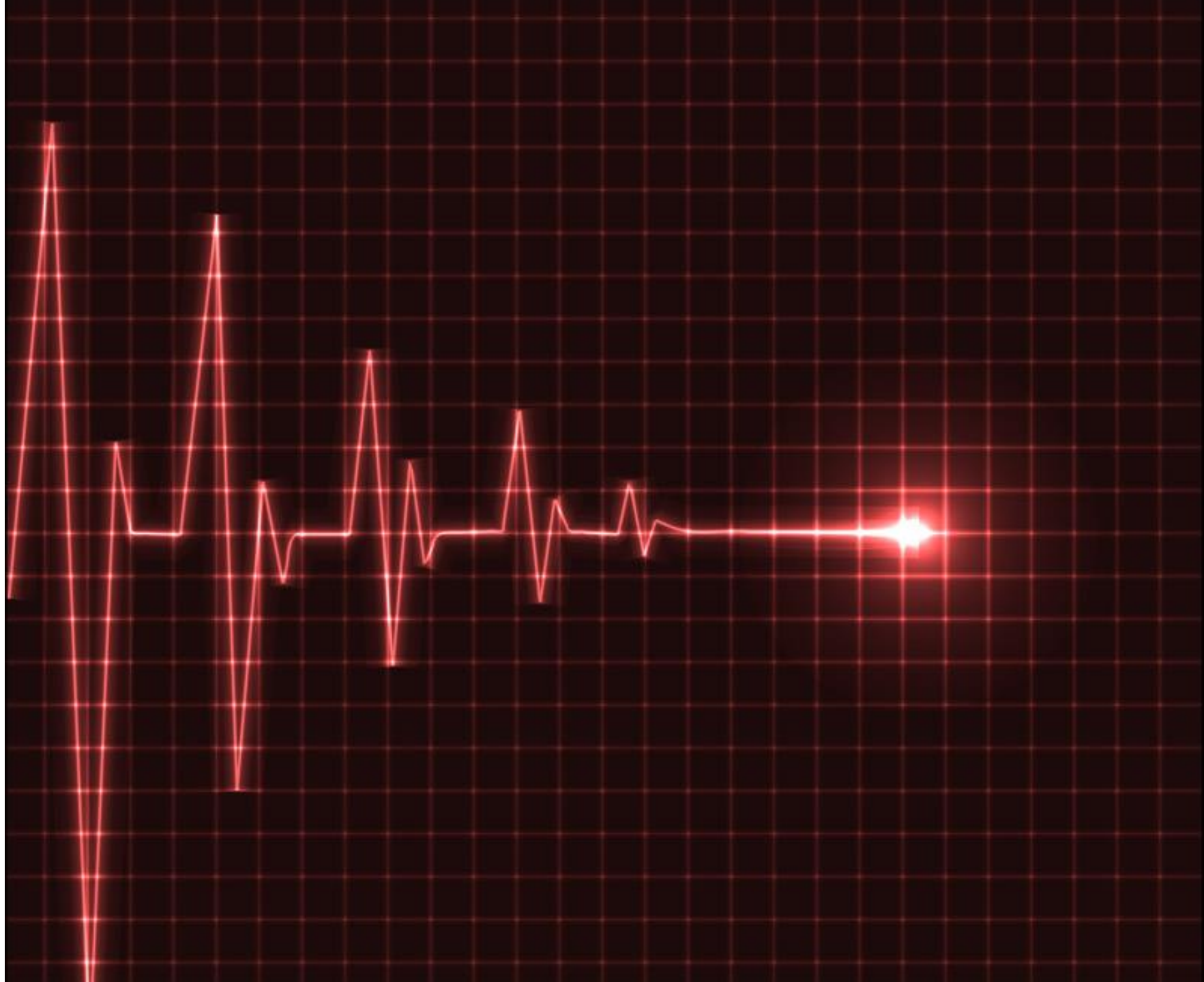
CVE-2017-3849

- Sending enrollee UDI as *space byte* or *null byte* crashes the registrar.
- No workaround for that, please upgrade your systems.



DeathKiss!

CVE-2017-3850



CVE-2017-3850

- The device is vulnerable even if the autonomic service is NOT enabled!
- Using 1st packet of adjacency discovery, with invalid TLVs crashes the device
- This attack can be launched remotely to crash the devices anywhere
- Block UDP for ports 8888, 4936.
- If you run AN then upgrade the software

Conclusion

- Autonomic Systems are smart systems that don't need human intervention to operate.
- Cisco AN protocol with its 3 phases has been reverse-engineered
- Cisco AN is vulnerable to:
 - CVE-2017-3849: crashing registrar with invalid UDIs
 - CVE-2017-3850: crashing IPv6 systems that supports AN
 - CVE-2017-6663: crashing the nodes by resetting secure channel multiple times
 - CVE-2017-6664: CRL on IOS XE not correctly implemented
 - CVE-2017-6665: denial-of-service for secure channel + Information disclosure

Finally...

- WireEdit 1.10.118 is the first application to support editing and the analyzing of the Cisco Autonomic Network protocol based on our analysis
- I would like to thank Marc Heuse for his contributions with protocol analysis
- 3-part series about Autonomic Network on insinuator.net
 - Introduction
 - Analysis
 - Vulnerabilities



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www.insinuator.net